

AWS VPC Transit Gateway & VPC Endpoint

1. Create one VPC, with 1 public subnet and 1 private subnet.

Step 1: Log in to AWS Management Console

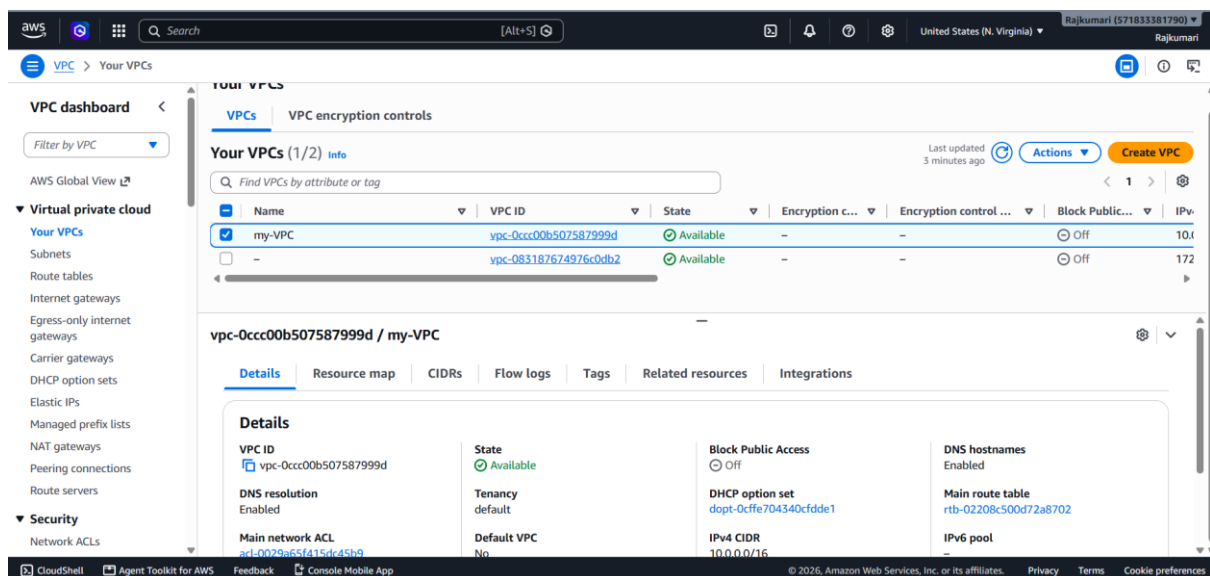
- Log in to the AWS Management Console.
- In the search bar, type **VPC**.
- Click **VPC** under **Services**.

Step 2: Open the VPC Dashboard

- In the left navigation pane, click **Your VPCs**.
- Click **Create VPC**.

Step 3: Create a New VPC

- Select **VPC only**.
- Enter the following details:
- Click **Create VPC**.



Step 4: Create a Public Subnet

- In the left pane, click **Subnets**.

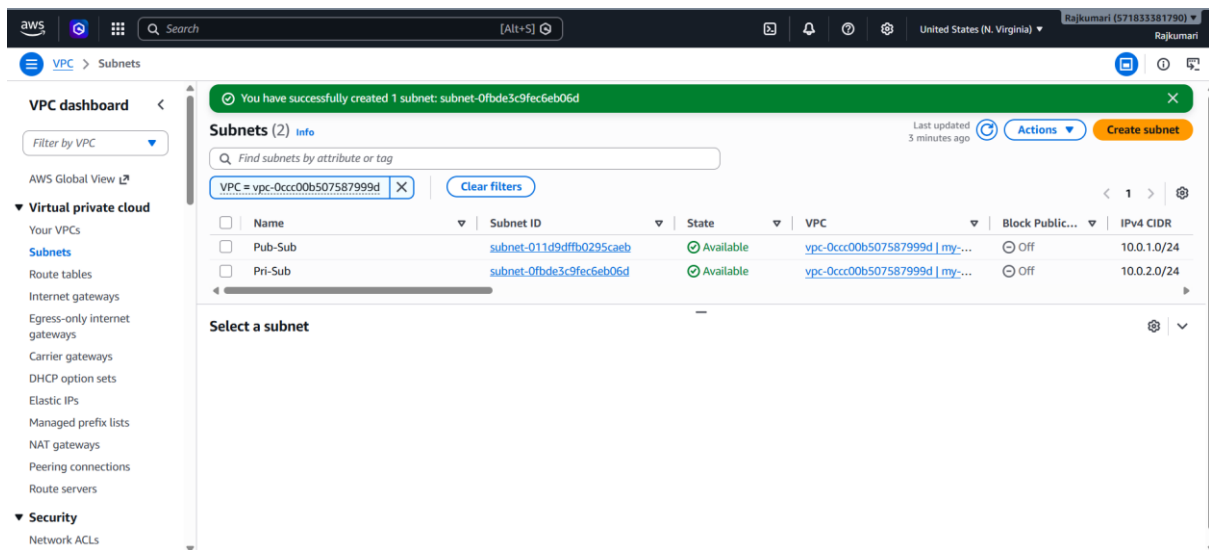
- Click **Create subnet**.

Enter the following information:

Click **Create subnet**.

Step 5: Create a Private Subnet

- Again click **Create subnet**.
- Click **Create subnet**.



Step 6: Enable Auto-Assign Public IP for Public Subnet

- Click **Subnets**.
- Select **PublicSubnet**.
- Choose the **Actions** menu.
- Click **Edit subnet settings**.
- Enable **Auto-assign public IPv4 address**.
- Click **Save**.

Expected Result:

- Instances launched in the public subnet automatically receive a public IP.

Step 7: Create an Internet Gateway

- In the left navigation pane, click **Internet Gateways**.

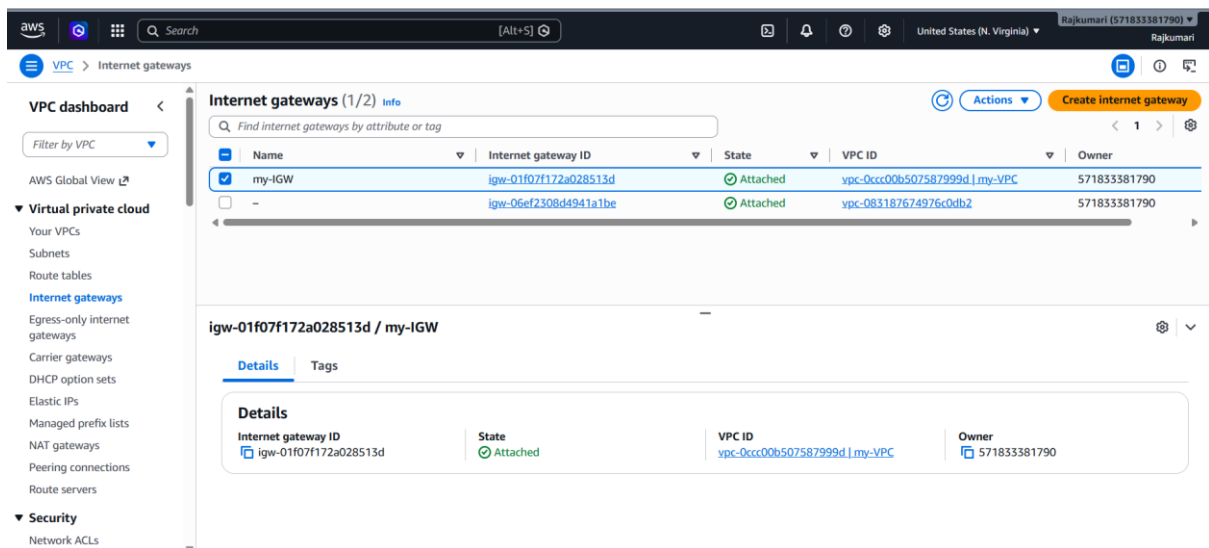
- Click **Create internet gateway**.

Step 8: Attach the Internet Gateway to the VPC

- Select **MyIGW**.
- Click **Actions**.
- Select **Attach to VPC**.
- Choose **MyVPC**.
- Click **Attach internet gateway**.

Expected Result:

- Internet Gateway is attached to the VPC.



Step 9: Create a Public Route Table

- Click **Route Tables**.
- Click **Create route table**.

Step 10: Add Internet Route

- Select **PublicRouteTable**.
- Open the **Routes** tab.
- Click **Edit routes**.
- Click **Add route**.

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

Step 11: Associate the Public Subnet with the Public Route Table

- Open **PublicRouteTable**.
- Go to the **Subnet Associations** tab.
- Click **Edit subnet associations**.
- Select **PublicSubnet**.
- Click **Save associations**.

Expected Result:

- Public subnet now has internet access.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ Pub-Sub	subnet-011d9dffb0295...	10.0.1.0/24	-	Main (rtb-02208c500d72a8702)
☐ Pri-Sub	subnet-0fbde3c9fec6eb...	10.0.2.0/24	-	Main (rtb-02208c500d72a8702)

Selected subnets

subnet-011d9dffb0295cae / Pub-Sub

Buttons: Cancel, Save associations

Step 12: Create a Private Route Table

- Click **Create route table**.
- Click **Create route table**.

Step 13: Associate the Private Subnet

- Select **PrivateRouteTable**.
- Open the **Subnet Associations** tab.
- Click **Edit subnet associations**.
- Select **PrivateSubnet**.
- Click **Save associations**.

The screenshot displays the AWS Management Console interface for the 'Route tables' section. A green notification banner at the top indicates a successful update of subnet associations for the 'Private-RT' route table. The main content area shows a list of route tables. The 'Private-RT' (ID: rtb-0fb822525e15963fe) is selected. Below this, the 'Subnet associations' tab is active, showing a single association for 'Pri-Sub' (Subnet ID: subnet-0fbde3c9fec6eb06d) with an IP v4 CIDR of 10.0.2.0/24.

2.Enable VPC peering for cross-region.

To complete the task "**Enable VPC Peering for cross-region**", you need **two VPCs in two different AWS Regions**. VPC peering cannot be configured with only one VPC.

Prerequisites: Create the following:

- **VPC 1**
 - Region: **ap-south-1 (Mumbai)**
 - CIDR: 10.0.0.0/16

- **VPC 2**
 - Region: **us-east-1 (N. Virginia)** (or any other AWS Region)
 - CIDR: 192.168.0.0/16 (must not overlap with VPC 1)

Step 1: Change the AWS Region

- Log in to the AWS Management Console.
- In the top-right corner, click the **Region** drop-down.
- Select **US West (Oregon) – us-west-2**.

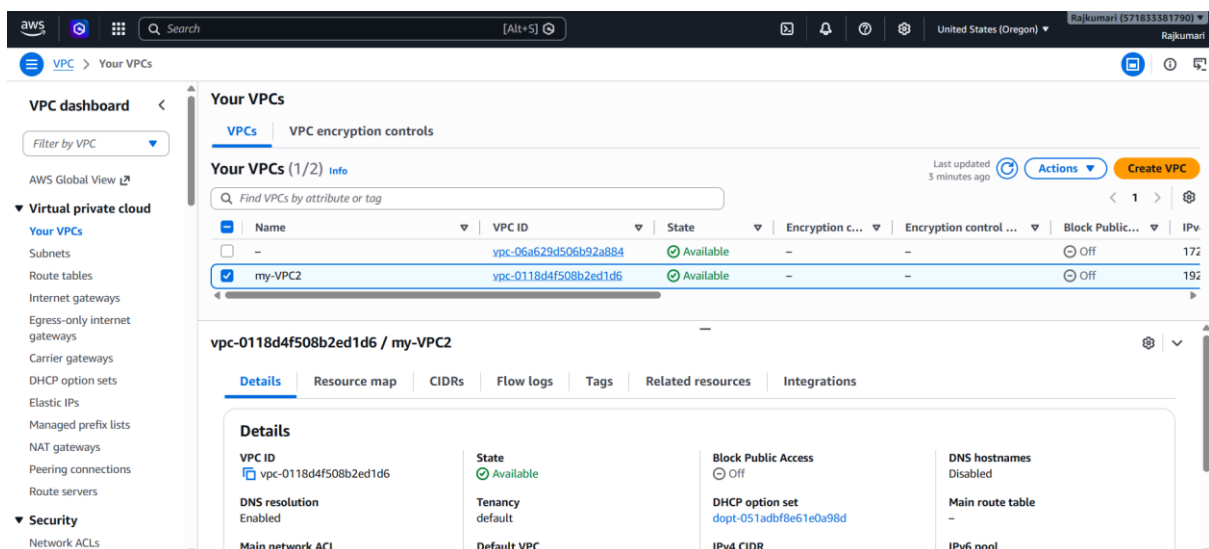
You can choose any AWS Region other than us-east-1.

Step 2: Open the VPC Dashboard

- In the search bar, type **VPC**.
- Click **VPC** under **Services**.
- In the left navigation pane, click **Your VPCs**.

Step 3: Create the Second VPC

- Click **Create VPC**.
- Select **VPC only**.



The screenshot shows the AWS Management Console VPC dashboard. The left navigation pane is open, showing 'Virtual private cloud' and 'Your VPCs'. The main content area displays a table of VPCs. The table has columns for Name, VPC ID, State, Encryption controls, Encryption control status, Block Public Access, and IPv4 CIDR. Two VPCs are listed: 'vpc-06a629d506b92a884' and 'my-VPC2' (vpc-0118d4f508b2ed1d6). The 'my-VPC2' VPC is selected, and its details are shown below the table. The details section includes tabs for Details, Resource map, CIDRs, Flow logs, Tags, Related resources, and Integrations. The 'Details' tab is active, showing the VPC ID, State (Available), Block Public Access (Off), DNS hostnames (Disabled), DHCP option set (dopt-051adbfb8e61e0a98d), Main network ACL, Tenancy (default), IPv4 CIDR, and IPv6 pool.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
-	vpc-06a629d506b92a884	Available	-	-	Off	172
my-VPC2	vpc-0118d4f508b2ed1d6	Available	-	-	Off	192

vpc-0118d4f508b2ed1d6 / my-VPC2

Details

VPC ID vpc-0118d4f508b2ed1d6	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-051adbfb8e61e0a98d	Main route table -
Main network ACL	Default VPC	IPv4 CIDR	IPv6 pool

Step 4: Create a Subnet in the Second VPC

- In the left pane, click **Subnets**.
- Click **Create subnet**.

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Pub-Sub2
The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (Oregon) / usw2-az2 (us-west-2a)

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/16

IPv4 subnet CIDR block
192.168.0.0/24 256 IPs

Tags - optional

Key	Value - optional	
Q, Name	Q, Pub-Sub2	Remove

Step 5: Switch Back to the First Region

- Click the **Region** drop-down.
- Select **US East (N. Virginia) – us-east-1**.

Step 6: Open Peering Connections

- In the VPC Console, click **Peering Connections**.
- Click **Create Peering Connection**.

The screenshot shows the AWS Management Console interface. At the top, the navigation bar includes the AWS logo, a search bar, and the user's profile (Rajkumari). The main content area is divided into two sections. The top section, titled 'Peering connection settings', shows the 'Create peering connection' wizard. It includes a 'Name' field with the value 'CrossRegPeering', a 'Select a local VPC to peer with' section with 'vpc-0ccc00b507587999d (my-VPC)' selected, and a 'Select another VPC to peer with' section with 'United States (Oregon) (us-west-2)' selected. The bottom section, titled 'VPC dashboard', shows a notification banner for a pending peering connection request 'pcx-07492664175bb776b / CrossRegPeering'. Below the notification, the 'Details' tab is active, showing information about the requester and acceptor VPCs, including their IDs, CIDRs, and regions. The 'DNS settings' section is also visible, showing that the requester VPC is configured to resolve DNS for private IP addresses.

Step 8: Accept the Peering Request

- Change the Region to **US West (Oregon)** – us-west-2.
- Open VPC.
- Click **Peering Connections**.
- Select the pending request.
- Click **Actions** → **Accept Request**.
- Click **Accept**.

Expected Result:

- The peering connection status changes to **Active**.

Peering connections (1/1) Info

Actions Create peering connection

View details Accept request Reject request Edit DNS settings Manage tags Delete peering connection

pcx-07492664175bb776b

Details DNS Route tables Tags

Details

Requester owner ID
571833381790

Peering connection ID
pcx-07492664175bb776b

Status

Peering connections (1/1) Info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC
-	pcx-07492664175bb776b	Active	vpc-0ccc00b507587999d

Step 9: Update the Route Table in the First VPC

- Switch back to US East (N. Virginia).
- Click **Route Tables**.
- Select the route table associated with your first VPC.
- Go to the **Routes** tab.
- Click **Edit routes**.

- Click **Add route**.

Edit routes

Route 1

Destination	Target	Status
10.0.0.0/16	local	Active

Propagated: No

Route Origin: CreateRouteTable

Route 2

Destination	Target	Status
192.168.0.0/16	Peering Connection	-

Propagated: No

Target dropdown suggestions:

- pcx-07492664175bb7 (selected)
- Use: "pcx-07492664175bb776b"
- pcx-07492664175bb776b (CrossRegPeering)

Remove

Add route

Step 10: Update the Route Table in the Second VPC

- Change to **US West (Oregon)**.
- Open **Route Tables**.
- Select the route table associated with **MyVPC2**.
- Go to the **Routes** tab.
- Click **Edit routes**.
- Click **Add route**.

The screenshot shows the AWS Management Console interface for editing routes in a VPC. The breadcrumb navigation indicates the path: VPC > Route tables > rtb-022977d86c37ad09f > Edit routes. The page title is 'Edit routes'.

Route 1

Destination	Target	Status
192.168.0.0/16	local	Active

Propagated: No
Route Origin: CreateRouteTable

Route 2

Destination	Target	Status
10.0.0.0/16	Peering Connection	-

Propagated: No

A dropdown menu for the target of Route 2 is open, showing the following options:

- pcx-07492664175bb7 (selected)
- Use: "pcx-07492664175bb776b"
- pcx-07492664175bb776b

A 'Remove' button is located at the bottom right of the route configuration area.

Step 11: Verify the Peering Connection

- In either Region, open **VPC** → **Peering Connections**.
- Verify that:
 - **Status:** Active
 - **Requester VPC:** Your VPC in us-east-1
 - **Accepter VPC:** MyVPC2 in us-west-2

3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Prerequisites

Your AWS Account (Account A)

- AWS Account A

- VPC1 created
- Example:
 - Region: us-east-1
 - CIDR: 10.0.0.0/16

Friend's AWS Account (Account B)

- AWS Account B
- VPC2 created
- Example:
 - Region: us-east-1 (same Region is easiest)
 - CIDR: 192.168.0.0/16 (must not overlap)

Step 1: Obtain Your Friend's AWS Account ID

Ask your friend to provide:

- AWS Account ID (12 digits)
- VPC ID
- Region

Step 2: Create the Peering Connection (Your Account)

- Log in to your AWS account.
- Open VPC.
- Click **Peering connections**.
- Click **Create peering connection**.

Fill in:

Field	Value
Name	CrossAccPeering
Requester VPC	Your VPC
Acceptor AWS Account	Another account

Field	Value
AWS Account ID	Your friend's 12-digit AWS Account ID
Region	us-east-1 (or your friend's Region)
Accepter VPC	Your friend's VPC

Click **Create peering connection**.

The first screenshot shows the 'Create peering connection' page in the AWS VPC console. It displays the 'VPC CIDRs for vpc-0ccc00b507587999d (my-VPC)' with a table showing the CIDR 10.0.0.0/16 is associated. Below this, the 'Select another VPC to peer with' section is visible, showing the 'Account' as 'Another account' with ID 479368491670, the 'Region' as 'Europe (Stockholm) (eu-north-1)', and the 'VPC ID (Accepter)' as vpc-0909c870a8c22eca5.

The second screenshot shows the 'pcx-010793be60d52f396 / CrossAccPeering' page. A green notification bar at the top states: 'A VPC peering connection pcx-010793be60d52f396 / CrossAccPeering has been requested. The owner of vpc-0909c870a8c22eca5 must accept the peering connection.' The page details the requester's VPC (vpc-0ccc00b507587999d / my-VPC) and the accepter's VPC (vpc-0909c870a8c22eca5). The status is 'Initiating Request to 479368491670' with an expiration time of Thursday, July 23, 2026 at 19:05:26 GMT+5:30. The DNS settings section shows 'Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses' is disabled.

Step 3: Friend Accepts the Request

Your friend should:

- Log in to their AWS account.
- Open **VPC → Peering connections**.

- Select the pending request.
- Click **Actions** → **Accept request**.

The status should change to **Active**.

The screenshot shows the AWS Management Console interface for VPC Peering connections. On the left, the navigation pane includes 'Virtual private cloud' and 'Security'. The main content area displays a table of peering connections. One connection, 'CrossAccPeering', is selected and highlighted. Below the table, the details for this connection are shown, including the requester and acceptor VPC IDs, their owner IDs, and the status 'Active'.

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
CrossRegPeering	pcx-07492664175bb776b	Active	vpc-0ccc00b507587999d / my-...	vpc-0118d4f508b2ed1d6
CrossAccPeering	pcx-010793be60d52f396	Active	vpc-0ccc00b507587999d / my-...	vpc-0909c870a8c22eca5

pcx-010793be60d52f396 / CrossAccPeering

Details

Requester owner ID 571833381790	Accepter owner ID 479368491670	VPC Peering connection ARN arn:aws:ec2:us-east-1:571833381790:vpc-peering-connection/pcx-010793be60d52f396
Peering connection ID pcx-010793be60d52f396	Requester VPC vpc-0ccc00b507587999d / my-VPC	
Status Active	Requester CIDRs 10.0.0.0/16	
		Accepter VPC vpc-0909c870a8c22eca5
		Accepter CIDRs ...

Step 4: Update Your Route Table

- Open **Route Tables**.
- Select the route table associated with your subnet.
- Click **Routes** → **Edit routes** → **Add route**.

Example:

Destination **Target**

192.168.0.0/16 VPC Peering Connection

The screenshot shows the AWS Management Console interface for editing a route table. The breadcrumb navigation indicates the path: VPC > Route tables > rtb-02208c500d72a8702 > Edit routes. The console displays two routes:

- Route 1:** Destination is 10.0.0.0/16, Target is 'local', and Status is 'Active'. The Route Origin is 'CreateRouteTable'.
- Route 2:** Destination is 192.168.0.0/16, Target is 'Peering Connection', and Status is 'Active'. A dropdown menu is open for the target, showing options: 'Use: "pcx-010793be60d52f396"', 'pcx-010793be60d52f396 (CrossAccPeering)', and 'pcx-07492664175bb776b (CrossRegPeering)'. There is a 'Remove' button next to the dropdown.

At the bottom, there is an 'Add route' button.

Step 5: Friend Updates Their Route Table

Your friend should add:

Destination Target

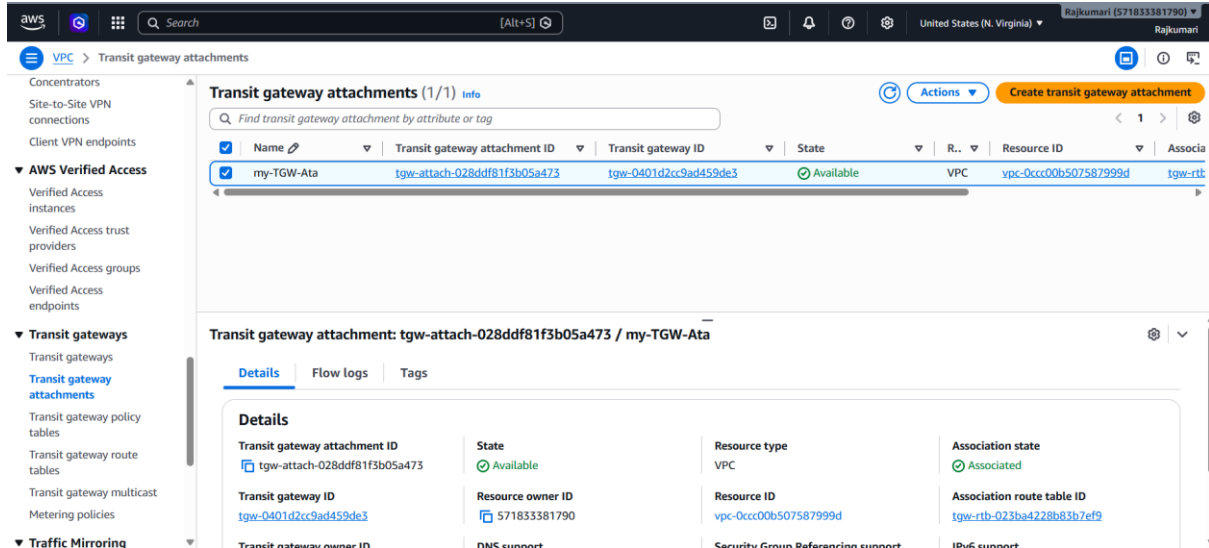
10.0.0.0/16 VPC Peering Connection (pcx-xxxxxx)

Save the changes.

Step 7: Test Connectivity (Optional)

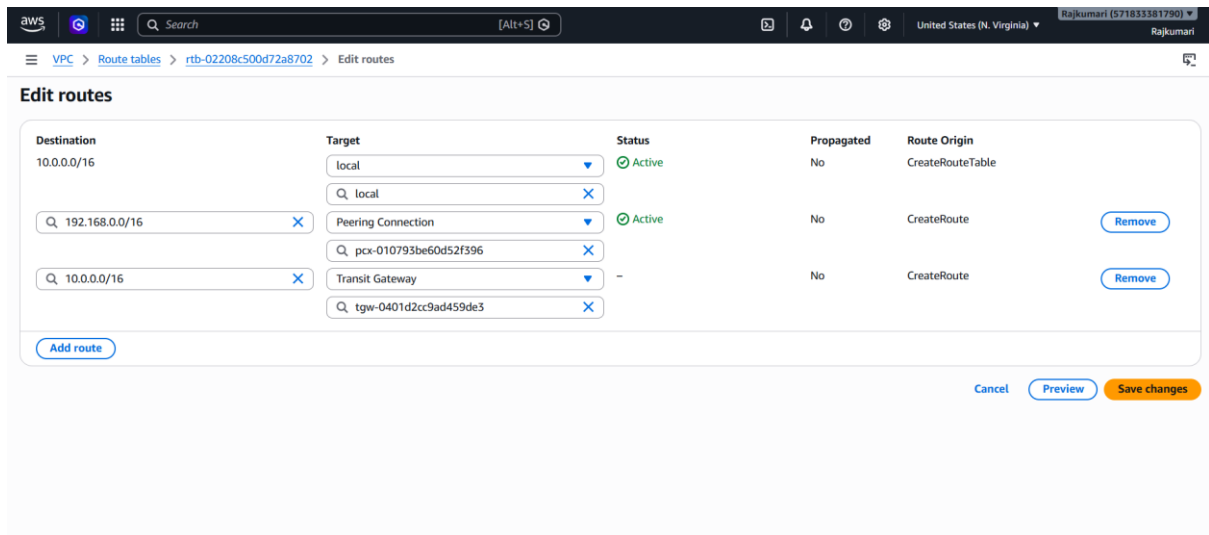
- Launch an EC2 instance in each VPC.
- Use the private IP addresses to test connectivity (for example, with ping if ICMP is allowed).

- Open **Transit Gateway Attachments**.
- Verify that both attachments show the status **Available**.



Step 8: Update the Route Table for VPC2

- Open **Route Tables**.
- Select the route table associated with **VPC2**.
- Click **Routes** → **Edit routes** → **Add route**.



5.Set up a VPC Endpoint.

Step 1: Log in to AWS Management Console

- Log in to the AWS Management Console.
- In the search bar, type **VPC**.
- Click **VPC** under **Services**.

Step 2: Open the Endpoints Page

- In the left navigation pane, scroll down to **Virtual private cloud**.
- Click **Endpoints**.

Step 3: Create a VPC Endpoint

- Click **Create endpoint**.

[VPC](#) > [Endpoints](#) > Create endpoint

Create endpoint Info

Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

Type Info

Select a category

☒ **AWS services**
Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint

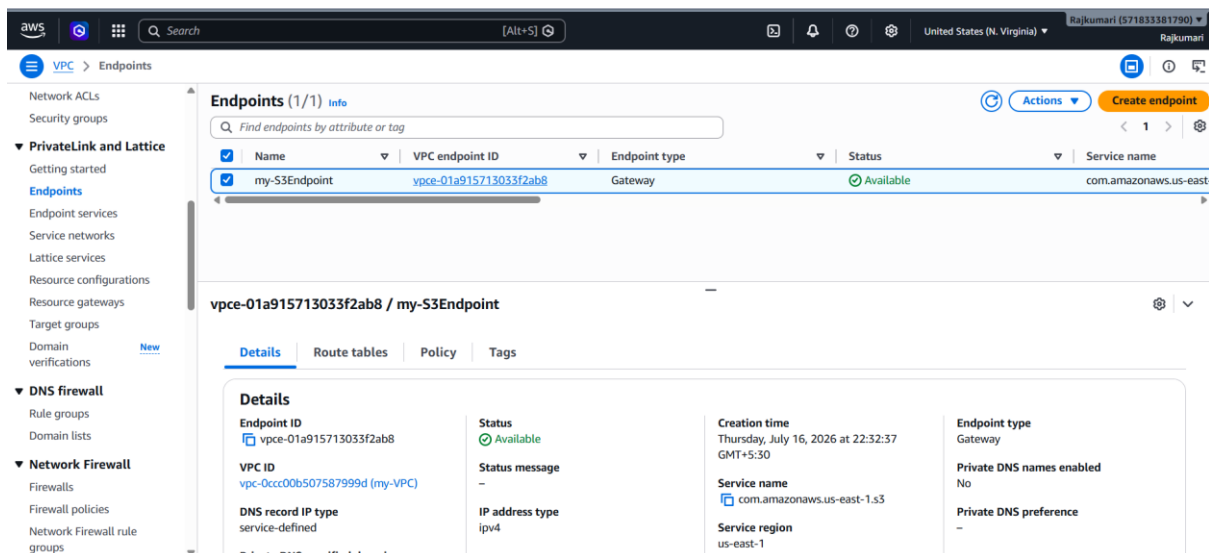
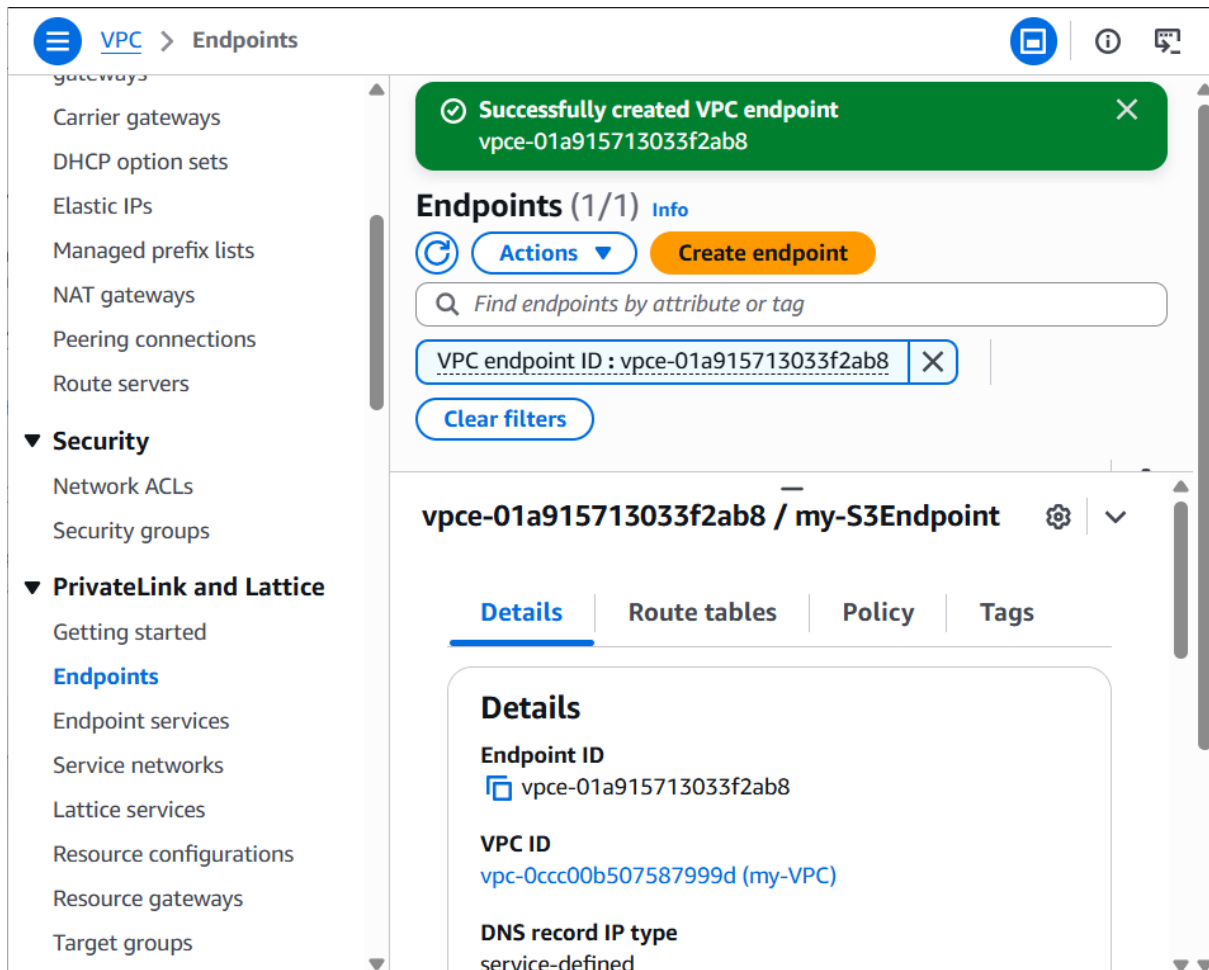
☐ **PrivateLink Ready partner services**
Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ **AWS Marketplace services**
Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint

☐ **EC2 Instance Connect Endpoint**
An elastic network interface that allows you to connect to resources in a private subnet

☐ **Resources**
Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink

☐ **Service networks**
Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink



Attach an IAM Role to the EC2 Instance

Step 1: Create an IAM Role

1. Open IAM → Roles.
2. Click **Create role**.

3. Select **AWS service**.
4. Choose **EC2**.
5. Click **Next**.
6. Attach a policy such as:
 - **AmazonS3ReadOnlyAccess** (to list buckets), or
 - **AmazonS3FullAccess** (for full access in a lab environment).
7. Give the role a name, for example:

EC2S3Role

8. Click **Create role**.

The screenshot shows the AWS IAM console interface. At the top, there's a navigation bar with the AWS logo, a search icon, and a user profile for 'Rajkumari (571833381790)'. Below the navigation bar, the breadcrumb trail reads 'IAM > Roles > Create role'. The main content area is titled 'Step 2: Add permissions' with an 'Edit' button. Under this title is a 'Permissions policy summary' table. The table has three columns: 'Policy name', 'Type', and 'Attached as'. It lists two policies: 'AmazonS3FilesFullAccess' and 'AmazonS3FilesReadOnlyAccess', both of type 'AWS managed' and attached as 'Permissions policy'. Below the table is 'Step 3: Add tags', which includes a section for 'Add tags - optional' with an 'Info' link. It explains that tags are key-value pairs for resource identification and notes that no tags are currently associated with the resource. An 'Add new tag' button is present, along with a note that up to 50 more tags can be added.

Policy name ↗ ▲	Type ▼	Attached as ▼
AmazonS3FilesFullAccess	AWS managed	Permissions policy
AmazonS3FilesReadOnlyAccess	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

The screenshot shows the AWS IAM console interface. On the left is a navigation sidebar for 'Identity and Access Management (IAM)' with links to Dashboard, Access Management (including Roles, Policies, IAM users, IAM user groups, Identity providers, Account settings, Root access management, and Temporary delegation requests), and Access reports (including Access Analyzer, Resource analysis, Unused access, and Analyzer settings). The main content area at the top features a green notification banner stating 'Role EC2S3Role created.' with a 'View role' button. Below this is the 'Roles (8)' section, which includes an 'Info' icon, 'Delete' and 'Create role' buttons, and a descriptive paragraph: 'An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.' A search bar and pagination controls (showing 1 of 8 items) are also present. At the bottom of the main area is a 'Roles Anywhere' section with an 'Info' icon, a 'Manage' button, and text explaining its purpose: 'Authenticate your non AWS workloads and securely provide access to AWS services.' and 'Access AWS from your non AWS workloads'.

Attach the Role to Your EC2 Instance

1. Open EC2 → Instances.
2. Select your EC2 instance.
3. Click **Actions** → **Security** → **Modify IAM role**.
4. Select EC2S3Role.
5. Click **Update IAM role**.

EC2

>

Instances

>

i-04cde4db9c806b788

>

Modify IAM role



Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID

 i-04cde4db9c806b788 (CrossAcc)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

EC2S3Role

▼



Create new IAM role

Cancel

Update IAM role

Launch an EC2 Instance

1. Open the **EC2** console.
2. Click **Launch Instance**.
3. Select:
 - **Amazon Linux 2023** (or Amazon Linux 2)
 - Your **VPC**
 - A subnet associated with the route table you selected for the VPC endpoint
4. Launch the instance.

Step 2: Connect to the EC2 Instance

From your local computer, connect to the EC2 instance.

Example:

```

brajk@RajKumari MINGW64 ~/Downloads
$ ssh -i linux-testing.pem ec2-user@98.92.34.148
#_
~\##### Amazon Linux 2023
~\#####\
~\###|
~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~V~'~>
~\
~\
~\m/'
Last login: Thu Jul 16 17:31:12 2026 from 103.248.210.235
[ec2-user@ip-10-0-1-182 ~]$ aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.18.36-69.138.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-1-182 ~]$ aws s3 ls
2026-07-15 18:35:16 vpc-flowlogs-bucket-raji
[ec2-user@ip-10-0-1-182 ~]$

```